



HASAN NASRALLAH

SENIOR AI ENGINEER

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01 · SUMMARY

AI Engineer with 7+ years of experience designing and deploying end to end machine learning systems, from ML ready datasets and model training to scalable cloud delivery. Currently focused on agentic AI: building autonomous agents that pair LLMs such as Claude by Anthropic with custom tools and MCP servers to run real world workflows. Specialized in Earth observation, remote sensing, and satellite imagery, translating complex geospatial data into impactful AI driven insights

02 · EXPERIENCE

Lead AI Engineer

RASID.AI · Lebanon

2023 → PRESENT

- ▶ Built Rasid GoPilot, an agentic AI framework for satellite imagery analysis that autonomously runs complete Earth observation workflows, built on MCP servers, custom tools, and AWS Bedrock, from scene selection and spectral analysis to file delivery
- ▶ Built and deployed models for urban building detection, urban change analysis, and general change detection using satellite imagery
- ▶ Led development of geospatial AI models for agriculture, including crop field delineation, wheat monitoring, and anomaly detection for crop health and stress analysis
- ▶ Built large scale datasets and trained models for global methane emission monitoring using real and synthetic satellite imagery, enabling detection and alerting workflows
- ▶ Designed and deployed a scalable Geo AI engine supporting segmentation, detection, classification, embeddings, and vision language models such as SAM and DINO
- ▶ Developed an automated Earth observation pipeline for multi source imagery ingestion across Sentinel 1, Sentinel 2, Landsat, and Mapbox

AI Engineer

Orbis Holding · Lebanon

2020 → 2023

- ▶ Developed and optimized computer vision models for face detection, tracking, recognition, emotion analysis, and liveness detection using CNNs, GANs, and advanced optimization in Python and C++
- ▶ Built AI powered OCR and document parsing systems for automated ID and passport verification in real world environments such as airports and retail
- ▶ Designed models for face image quality assessment, blur detection, and forgery detection in identity documents to strengthen verification accuracy and security

Research Assistant

National Council for Remote Sensing, Lebanon (CNRS)

2018 → 2020

- ▶ Developed end to end AI pipelines for building detection from satellite imagery, covering ML ready dataset creation, model design, training, and large scale deployment
- ▶ Generated a nationwide urban map of Lebanon and designed a solar energy estimation algorithm to evaluate photovoltaic potential across buildings
- ▶ Built a large scale dataset and trained deep learning models for silo bin detection from Mapbox satellite imagery

03 · EDUCATION

Masters in Computer and Communication Engineering

Lebanese University

2015 → 2020 · Beirut, Lebanon

7+

YEARS EXP

4

PAPERS

20+

PROJECTS

RASID GOPILLOT

SEEING EARTH, SMARTER

Flagship product I lead at RASID: an AI agent for geospatial intelligence powered by MCP servers, custom tools, and AWS Bedrock, spanning 6 satellite sources, 17+ spectral indices, and 10+ GPU deep learning models.

▶ app.rasid.ai/try-gopilot

SKILLS

MACHINE LEARNING

Computer Vision · Deep Learning · PyTorch · AWS · GCP · Vision Language Models · Foundation Models · Agents · MLOps · LLMS

PROGRAMMING

Python · C/C++ · Java

REMOTE SENSING

QGIS · ArcGIS · Google Earth Engine · GDAL

KEY ACHIEVEMENTS

◆ ML Competitions

Developed solutions for NASA Harvest, SpaceNet, Open Cities, Xview2, and BirdCleff 2022

◆ Published Researcher

Author of multiple peer reviewed papers in geospatial AI

PUBLICATIONS

[Zero Shot Refinement of Buildings Segmentation Models using SAM](#)

ARXIV

[Empirical Study of PEFT Techniques for Winter Wheat Segmentation](#)

ARXIV

[Scale Invariant Model for Building Detection from Aerial Images](#)

SPRINGER

[Lebanon Solar Rooftop Potential Assessment via Buildings Segmentation](#)

IEEE